

Question	Marking Guidance	Mark	Comments
03.1	1. Tip produces IAA; 2. Affects concentration of IAA OR Affects (shoot) length/growth/elongation; 3. Mitosis/division occurs in shoot tips; 4. Affects (shoot) length/growth/elongation;	2 max	Mark in pairs 1 and 2 or 3 and 4. 1 and 2. Accept auxin for IAA. 1. Ignore contains/stores IAA. 1. Accept affects amount of IAA. 2. Accept affects independent variable. 2 and 4. Ignore affects results.
03.2	1. For respiration; 2. Provide ATP/energy (for growth);	2	1. Ignore photosynthesis. 1. Ignore aerobic/anaerobic (respiration). 1. Reject glucose used in photosynthesis. 2. Reject produce energy. 2. Do not credit photosynthesis provides ATP.
03.3	1. To prevent/reduce evaporation; 2. (Which) alters concentration of (IAA) solution OR (Which) alters water potential;	2	1. Accept evaporation of (IAA/glucose) 'solution'. 2. Accept auxin for IAA. 1. Ignore contamination.
03.4	1. Increase in IAA <u>concentration</u> the higher/greater the <u>mean</u> (change in) length; 2. (High) IAA stimulates <u>cell elongation</u> ; 3. In roots, growth/elongation less/inhibited;	3	1, 2 and 3. Accept auxin for IAA. 3. Accept decrease in (mean) change in length but reject 'decreases length' on its own.

			3. Accept 'opposite results or 'negative correlation'.
03.5	0.4 and 39.6;	1	Both numbers required and must be in order shown.
